

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 70% Ethanol

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>R40135, R2470110</b>                                                                              |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
Flammable liquids

Category 2

**Health hazards**

Specific target organ toxicity - (single exposure)

Category 2

**Environmental hazards**  
No hazards identified

### Label Elements



Flame



Health Hazard

**Signal Word**

**Danger**

**Hazard Statements**

H225 - Highly flammable liquid and vapor

H371 - May cause damage to organs

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P235 - Store in a well-ventilated place. Keep cool

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

This product does not contain any known or suspected endocrine disruptors

**Section 3 - Composition and Information on Ingredients**

| Component      | CAS No  | Weight % |
|----------------|---------|----------|
| Ethyl alcohol  | 64-17-5 | 66.5     |
| Methyl alcohol | 67-56-1 | 3.5      |

**Section 4 - First Aid Measures**

|                                            |                                                                                                                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                      |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                 |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                   |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                  |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                   |

**Section 5 - Fire Fighting Measures**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Hazardous Decomposition Products**

Carbon oxides.

**Specific Hazards Arising from the Chemical**

Flammable. Keep product and empty container away from heat and sources of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Combustible material.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep away from heat, sparks and flame. Keep containers tightly closed in a cool, well-ventilated place. Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component      | Australia                                                                                  | New Zealand WEL                                                                                    | ACGIH TLV                             | The United Kingdom                                                                                               | Germany                                                           |
|----------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Ethyl alcohol  | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                               | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                                       | STEL: 1000 ppm                        | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL; 5760 mg/m <sup>3</sup><br>STEL | 200 ppm TWA MAK;<br>380 mg/m <sup>3</sup> TWA MAK                 |
| Methyl alcohol | STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup> | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | WEL - TWA: 200 ppm<br>TWA; 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL; 333 mg/m <sup>3</sup> STEL  | 100 ppm TWA MAK;<br>130 mg/m <sup>3</sup> TWA<br>MAKSkin absorber |

### Biological limit values

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component      | Australia | New Zealand                                   | European Union | United Kingdom | Germany                                                                                                                                     |
|----------------|-----------|-----------------------------------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol |           | 15 mg/L (urine) end of shift (Methyl alcohol) |                |                | Methanol: 15 mg/L urine (end of shift )<br>Methanol: 15 mg/L urine (for long-term exposures: at the end of the shift after several shifts ) |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                 |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.                                                                                                                                                                                                                       |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                                           |                                                                                       |
|------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Appearance</b>                              | Clear Colourless                                          |                                                                                       |
| <b>Physical State</b>                          | Liquid                                                    |                                                                                       |
| <b>Odor</b>                                    | Alcohol-like                                              |                                                                                       |
| <b>Odor Threshold</b>                          | No data available                                         |                                                                                       |
| <b>pH</b>                                      | No information available                                  |                                                                                       |
| <b>Melting Point/Range</b>                     | No data available                                         |                                                                                       |
| <b>Softening Point</b>                         | No data available                                         |                                                                                       |
| <b>Boiling Point/Range</b>                     | No information available                                  |                                                                                       |
| <b>Flash Point</b>                             | 65 °C / 149 °F                                            | <b>Method -</b> CC (closed cup)                                                       |
| <b>Evaporation Rate</b>                        | No data available                                         |                                                                                       |
| <b>Flammability (solid,gas)</b>                | Not applicable                                            | Liquid                                                                                |
| <b>Explosion Limits</b>                        | No data available <b>Lower</b> 3.3%<br><b>Upper</b> 19.0% |                                                                                       |
| <b>Vapor Pressure</b>                          | No data available                                         |                                                                                       |
| <b>Vapor Density</b>                           | No data available                                         | (Air = 1.0)                                                                           |
| <b>Specific Gravity / Density</b>              | No data available                                         |                                                                                       |
| <b>Bulk Density</b>                            | Not applicable                                            | Liquid                                                                                |
| <b>Water Solubility</b>                        | No information available                                  |                                                                                       |
| <b>Solubility in other solvents</b>            | No information available                                  |                                                                                       |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                           |                                                                                       |
| <b>Component</b>                               | <b>log Pow</b>                                            |                                                                                       |
| Ethyl alcohol                                  | -0.32                                                     |                                                                                       |
| Methyl alcohol                                 | -0.74                                                     |                                                                                       |
| <b>Autoignition Temperature</b>                | 363 °C / 685.4 °F                                         |                                                                                       |
| <b>Decomposition Temperature</b>               | No data available                                         |                                                                                       |
| <b>Viscosity</b>                               | No data available                                         |                                                                                       |
| <b>Explosive Properties</b>                    |                                                           | Vapors may form explosive mixtures with air<br>explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>                    | No information available                                  |                                                                                       |

### Other information

## Section 10 - Stability and Reactivity

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Reactivity</b> | None known, based on information available   |
| <b>Stability</b>  | Stable under recommended storage conditions. |

|                                         |                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Heat, flames and sparks, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | None known.                                                                                |
| <b>Hazardous Decomposition Products</b> | Carbon oxides.                                                                             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                   |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|                   |                   |
|-------------------|-------------------|
| <b>Oral</b>       | No data available |
| <b>Dermal</b>     | No data available |
| <b>Inhalation</b> | No data available |

#### Toxicology data for the components

| Component      | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|----------------|--------------------------------|-------------------------------|-------------------------------|
| Ethyl alcohol  | LD50 = 7060 mg/kg ( Rat )      |                               | 20000 ppm/10H ( Rat )         |
| Methyl alcohol | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

##### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

| Component                         | Test method                                                       | Test species | Study result    |
|-----------------------------------|-------------------------------------------------------------------|--------------|-----------------|
| Methyl alcohol<br>67-56-1 ( 3.5 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available

| Component                         | Test method             | Test species / Duration       | Study result           |
|-----------------------------------|-------------------------|-------------------------------|------------------------|
| Methyl alcohol<br>67-56-1 ( 3.5 ) | OECD Test Guideline 416 | Rat / Inhalation 2 Generation | NOAEC = 1.3 mg/l (air) |

**(h) STOT-single exposure;** No data available

|                                |                                             |
|--------------------------------|---------------------------------------------|
| <b>Results / Target organs</b> | Optic nerve<br>Central nervous system (CNS) |
|--------------------------------|---------------------------------------------|

**(i) STOT-repeated exposure;** No data available

|                      |                           |
|----------------------|---------------------------|
| <b>Target Organs</b> | No information available. |
|----------------------|---------------------------|

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

### Ecotoxicity effects

| Component      | Freshwater Fish                                                  | Water Flea                                    | Freshwater Algae                              | Microtox                                                                                                                |
|----------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol  | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |
| Methyl alcohol | Pimephales promelas:<br>LC50 > 10000 mg/L 96h                    | EC50 > 10000 mg/L 24h                         |                                               | EC50 = 39000 mg/L 25<br>min<br>EC50 = 40000 mg/L 15<br>min<br>EC50 = 43000 mg/L 5<br>min                                |

**Persistence and Degradability** Expected to be biodegradable

**Persistence** Soluble in water.

| Component                       | Degradability                  |
|---------------------------------|--------------------------------|
| Methyl alcohol<br>67-56-1 (3.5) | DT50 ~ 17.2d<br>>94% after 20d |

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| Ethyl alcohol  | -0.32   | No data available             |
| Methyl alcohol | -0.74   | <10 dimensionless             |

**Mobility** No information available.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

### IMDG/IMO

|                      |                  |
|----------------------|------------------|
| UN-No                | UN1170           |
| Proper Shipping Name | Ethanol solution |
| Hazard Class         | 3                |
| Packing Group        | II               |

**ADG**

UN-No UN1170  
 Proper Shipping Name Ethanol solution  
 Hazard Class 3  
 Packing Group II

| Component                         | Hazchem Code |
|-----------------------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 66.5 ) | 2YE<br>2Y    |
| Methyl alcohol<br>67-56-1 ( 3.5 ) | 2WE          |

**IATA**

UN-No UN1170  
 Proper Shipping Name Ethanol solution  
 Hazard Class 3  
 Packing Group II

Environmental hazards No hazards identified  
 Special Precautions No special precautions required  
 Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol - 67-56-1 | Schedule 5 listed - except its derivatives; in preparations except a) when included in Schedule 10, or b) in preparations containing ≤2% of Methanol, or c) when Methanol is present only as a denaturant of Ethanol<br>Schedule 6 listed - except its derivatives; except a) when included in Schedule 5, or b) when included in Schedule 10, or c) in preparations containing ≤2% of Methanol<br>Schedule 10 listed |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------|-------------------------------------------------------------|------------------------|
| Ethyl alcohol - 64-17-5  | Present                                                     | -                      |
| Methyl alcohol - 67-56-1 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security



Concern

National pollutant inventory Subject to reporting requirements

| Component                | National pollutant inventory      |
|--------------------------|-----------------------------------|
| Ethyl alcohol - 64-17-5  | 10 tonne/yr. Threshold category 1 |
| Methyl alcohol - 67-56-1 | 10 tonne/yr. Threshold category 1 |

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component      | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Ethyl alcohol  | X    | X     | 200-578-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-13217 |
| Methyl alcohol | X    | X     | 200-659-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-23193 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**International Regulations****Ozone Depletion Potential** This product does not contain any known or suspected substance**Persistent Organic Pollutant** This product does not contain any known or suspected substance**Rotterdam Convention (PIC)** Not applicable**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component               | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-------------------------|------------------------------------|------------------------------------------------------------------------|
| Ethyl alcohol - 64-17-5 | Annex I - Y42                      | Y42 except Halogenated solvents                                        |

| Component      | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethyl alcohol  | 64-17-5 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Methyl alcohol | 67-56-1 | Listed   | Not applicable                             | 500 tonne                                                                                 | 5000 tonne                                                                               |

**Authorisation/Restrictions according to EU REACH**

| Component      | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl alcohol | -                                                                   | Use restricted. See item 69.<br>(see link for restriction details)<br>Use restricted. See item 75. | -                                                                                                     |

|  |  |                                    |  |
|--|--|------------------------------------|--|
|  |  | (see link for restriction details) |  |
|--|--|------------------------------------|--|

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

|                         |                       |
|-------------------------|-----------------------|
| <b>Revision Date</b>    | 30-Jun-2023           |
| <b>Revision Summary</b> | Update to CLP Format. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**